



Chimeric Forecasting: An experiment to leverage human judgment to improve forecasts of infectious disease using simulated surveillance data

Thomas McAndrew^{a,*}, Graham C. Gibson^b, David Braun^c, Abhishek Srivastava^d, Kate Brown^a

^a Department of Community and Population Health, College of Health, Lehigh University, Bethlehem PA, United States of America

^b Statistical Sciences, Los Alamos National Laboratory, Los Alamos, NM, United States of America

^c Department of Psychology College of Arts and Science, Lehigh University, Bethlehem PA, United States of America

^d P.C. Rossin College of Engineering & Applied Science, Lehigh University, Bethlehem PA, United States of America

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ABSTRACT

Forecasts of infectious agents provide public health officials advanced warning about the intensity and timing of the spread of disease. Past work has found that accuracy and calibration of forecasts is weakest when attempting to predict an epidemic peak. Forecasts from a mechanistic model would be improved if there existed accurate information about the timing and intensity of an epidemic. We presented 3000 humans with simulated surveillance data about the number of incident hospitalizations from a current and two past seasons, and asked that they predict the peak time and intensity of the underlying epidemic. We found that in comparison to two control models, a model including human judgment produced more accurate forecasts of peak time and intensity of hospitalizations during an epidemic. Chimeric models have the potential to improve our ability to predict targets of public health interest which may in turn reduce infectious disease burden.

1. Introduction

Forecasts of the trajectory of a pathogen provide public health decision makers advanced warning of the potential timing and intensity of an epidemic or pandemic event so that they may take actions to mitigate the spread of disease and potentially reduce disease burden (Rivers et al., 2019; Lutz et al., 2019; Biggerstaff et al., 2022; Reich et al., 2019; Viboud et al., 2018). The significance of infectious diseases forecasts has been highlighted by public health practitioners and the dramatic growth of forecast challenges and research into ensemble forecasts for a diverse number of infectious agents (Lutz et al., 2019; Reich et al., 2019; Viboud et al., 2018; Oidtmann et al., 2021).

Computational forecasts of an infectious disease map observed data from a surveillance system to a sequence of point predictions, full predictive densities, or summaries of a predictive density over a future epidemiological target such as one through four week ahead forecasts of incident hospitalizations. Models that generate forecasts can be broadly classified as either phenomenological or mechanistic. Phenomenological models leverage correlations among several different sources of data—including past observations of the infectious disease of interest—to describe how an infectious disease evolves over time (Pell et al., 2018; Chowell et al., 2016). Mechanistic models suppose that at a given point in time individuals can be categorized into one of a finite number of states that represent their disease status and that there

exists a system of equations to describe the dynamics of these states over time (Cobey, 2020; Bjørnstad et al., 2020). Computational models of infectious diseases often augment surveillance data with signals such as internet search data, genetic information of the pathogen, meteorological data, and mobility data (Buckee et al., 2020; Santillana et al., 2015; Lemey et al., 2014; Shaman and Kohn, 2009; George et al., 2019). Forecast “hubs” are entities that act to organize, store, and combine many forecasts from computational models into a single, cohesive statement about the potential future of a pathogen (Cramer et al., 2022; Reich et al., 2019).

An alternative to computational models to describe the path of an infectious agent are crowdsourced models of human judgment forecasts (Farrow et al., 2017; McAndrew et al., 2022c; Venkatramanan et al., 2022; Braun et al., 2022). Human judgment forecasts collect from individuals either point predictions or predictive densities and aggregate these individuals’ predictions into a single ensemble forecast of some future target of interest such as influenza-like illness, incident cases and deaths due to COVID-19, or number of cases of monkeypox (Farrow et al., 2017; McAndrew et al., 2022c; Codi et al., 2022). Past work in human judgment forecasting for infectious diseases has at times shown improved performance compared to computational models, can be faster to implement than designing a computational model, and humans have the ability to answer questions that may be difficult

* Corresponding author.

E-mail address: mcandrew@lehigh.edu (T. McAndrew).

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for a computational model to answer (Codi et al., 2022; McAndrew and Reich, 2022; McAndrew et al., 2022a; Bosse et al., 2022). However there too are disadvantages to applying human judgment to infectious disease forecasting. Humans are prone to overconfidence, biases such as anchoring, and can too heavily rely on heuristic information (Recchia et al., 2021; Lawrence and O'Connor, 1992; Tversky and Kahneman, 1974).

Here we present work on chimeric forecasting, forecasts produced by a model that integrates observed data from a surveillance system and predictions generated by a generalist crowd of humans (Fig. 1). The goal of this model is to forecast the number of incident hospitalizations over time, which were generated by a simulation, by learning from two sources of information: (i) daily incident hospitalizations reported by a surveillance system and (ii) crowdsourced human judgment predictions of the day when the number of incident hospitalizations will be highest (the peak time) and the number of hospitalizations that will be reported on that day (peak intensity).

We hypothesize that a chimeric mechanistic model can leverage both objective data from a surveillance system and subjective predictions generated by humans to produce forecasts of the number of incident hospitalizations due to a simulated pathogen which have improved performance for predictions of peak time and peak intensity compared to mechanistic models that do not incorporate human judgment (see supplemental Figs. S1 and S2 that present calibration results as the impetus for this work).

To test our hypothesis, we simulated a fictitious epidemic over three seasons—two past seasons and a current, in progress, season—and represented a surveillance system as a sequence of noisy observations centered around the true number of incident hospitalizations (The experimental setup is detailed in the methods section below). We recruited 3000 humans from Prolific—a crowdsourcing platform—to participate in this experiment and asked that they first observe a plot of simulated surveillance data and second predict the peak time and peak intensity of the current season underlying epidemic that the surveillance system is recording (The survey is provided in the supplemental information). Humans were randomized to one of 30 different arms that varied the number of days of data for the current season that were recorded by the surveillance system (4 weeks before the peak, 3, 2, 1 week before and 2 weeks after the peak), level of noise (low, medium, high) in the surveillance system, and whether or not humans were presented with a model that was trained on current season surveillance data to help guide their prediction.

Our chimeric model is a Bayesian SEIHR dynamical system which uses a ‘Bayesian melding’ approach to combine two sources of information about the system (Poole and Raftery, 2000) (see supplemental section titled “Problem formulation and models” for a detailed description of the modeling approach). One set of prior distributions guide traditional epidemiological parameters in the system such as the initial proportion of susceptible, exposed, and infectious individuals, parameters that control how disease states evolve over time such as R_0 , and the level of noise that is present in the surveillance system. However, our innovation in this work is to notice that prior densities over the first set of epidemiological parameters in the system also induce prior densities over the peak time and intensity of a potential epidemic. Because we can generate a density over peak time and intensity, we can include subjective information about the peak time and intensity from human judgment predictions of these quantities (see figures S3 and S4 for predictive densities over peak time and intensity that were generated via human judgment).

By adding a prior density over the peak time and intensity we allow our model to learn from humans judgment predictions over a lower dimensional space, which we expect to be an easier task for humans to complete compared to producing predictions over the entire time series of incident hospitalizations. See methods section “Chimeric mechanistic model” and Figure S5 for a presentation of the chimeric model.

2. Methods

2.1. Survey, recruitment, and experimental setup

Crowdsourcing was conducted on the platform Prolific and the survey that was distributed to humans was created using the Qualtrics platform (Anon, 0000b; Palan and Schitter, 2018). Prolific is a crowdsourcing platform that allows any user to volunteer their time to complete tasks for a monetary reward (Palan and Schitter, 2018). We paid volunteers who participated in this study \$1.00 USD. Participants were those who had accounts on the crowdsourcing platform Prolific (Palan and Schitter, 2018). Those who could participate were required to reside in the US, and complete the study using either a computer or tablet (mobile was not allowed). In addition, participants were required to be fluent in English. To note, there were not conditions on skill or expertise as it relates to infectious disease epidemiology.

Our crowdsourcing experiment presented humans with a figure that displayed the simulated number of incident hospitalizations collected by our fictitious surveillance system for the current season from day one of the epidemic up until a time, T , that was less than 210 days (30 weeks). In addition to the current season trajectory of incident hospitalizations, we included two additional trajectories for all 210 days of incident hospitalizations from past seasons that had different values of R_0 . The R_0 for the current season was set to 1.75. The R_0 for the past two seasons were drawn at random as 1.75 plus a uniform random variable with lower bound $-1/2$ and upper bound $1/2$, or $R_0^s = 1.75 + U(-1/2, 1/2)$. The R_0 values for the two past seasons remained identical across treatments and were equal to $R_0 = 2.22$ for the past season and $R_0 = 1.46$ for the other. We decided to keep the three underlying mechanistic models that were used to generate noisy observations constant throughout this experiment to make predictions by humans and models comparable to one another.

We asked humans to read and sign a consent document, read text that described a figure that displayed realizations of incident hospitalizations from a noisy surveillance system as blue circles, noisy realizations of a past season in gray and noisy realizations of a second past season in black. The graph was produced using plotly which allows a user to interact with the graph (Anon, 0000a). An example graph is presented in Fig. 1 and the true underlying curves are presented in Fig. 2. Half of the humans were randomized to treatments that presented them with a model that was trained on current season surveillance data and produced a forecast from time T to time 210 on top of surveillance data in the figure. This model was a SEIRH model trained on the current season data only (i.e. blue dots in the figure that were presented) and we presented in the figure in red a median, 50 percent prediction interval, and 95 percent prediction interval.

All humans were presented with the same prompt and task text. The task that we assigned asked that a crowd member input two pieces of information: (i) the day that they predict the number of hospitalizations will be highest (peak time) and (ii) the number of incident hospitalizations that will be reported by the surveillance system on this day (peak intensity). After humans made predictions for the peak day and peak intensity they were asked to assess what type of information they relied on to produce these predictions, and, as a measure of their understanding of the task, they were asked to paraphrase the assignment in no more than two sentences. We did not describe to humans: the total population, that the model presented (if applicable) was used to simulate incident hospitalizations.

The entire survey in pdf format, which includes all 30 potential treatments to randomize a human to, is attached to this work as supplementary material.

Humans were enrolled randomly into one of 30 treatments based on (i) the length of time they observed the surveillance system, (ii) the amount of noise present in the surveillance system, and (iii) whether they were presented with a model that estimated the trajectory of

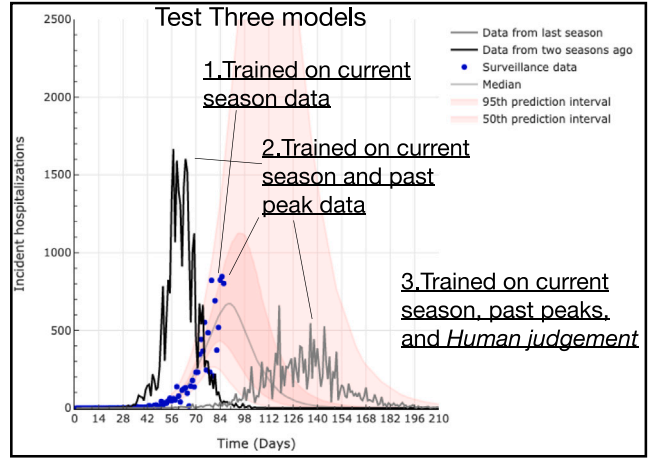
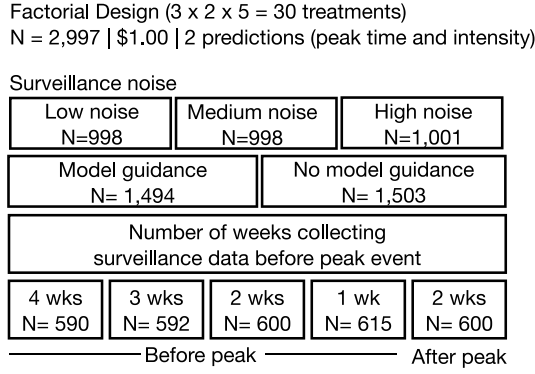


Fig. 1. Summary of experiment and example figure presented to humans. (Left) This crowdsourcing experiment randomized 2997 participants into one of 30 factors that depended on: the level of noise in the surveillance system, whether participants were presented with a model or not, and the number of weeks of surveillance data presented before (or after) the peak. (Right) Example of noisy surveillance data for the current season represented as blue circles and past seasons represented as gray and black lines plus a model forecast in red that was presented to crowd members. Humans were asked to use this data to predict the peak time and intensity. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1
Number of participants from Prolific who were randomized to one of thirty treatments.

	4 wks before	3 wks	2 wks	1 wk	2 wks after
High noise					
Model guidance	102	103	101	101	94
No model guidance	98	96	105	102	99
Medium noise					
Model guidance	94	102	96	107	104
No model guidance	100	93	100	101	101
Low noise					
Model guidance	94	101	99	99	97
No model guidance	102	97	99	105	105

hospitalizations. Counts of the number of participants who were randomized to each treatment can be found in Table 1.

Humans were randomized into one of five different lengths of time that the surveillance system had collected data for the current season: four weeks before the (unknown to the participant) peak, 3 weeks before the peak, 2 weeks before the peak, 1 week before the peak, and 2 weeks after the peak. For each of these weeks humans were randomized to one of three levels of noise in the surveillance system: low noise that corresponded to a standard deviation of 96 hospitalizations if the true number of incident hospitalizations at the peak was 300 (we used the negative binomial 2, NB2(300,10), function from numpyro), medium noise that corresponded to a standard deviation of 135 hospitalizations if the true number of incident hospitalizations at the peak was 300 (NB2(300,5)), and high noise with a standard deviation of 190 hospitalizations if the true number of incident hospitalizations at the peak was 300 (NB2(300, 2.50)). Finally, humans were randomized to be presented with or without a forecast from a SEIHR mechanistic model trained on the observed surveillance system data. From the model, we presented humans with the daily median, 25th and 75th percentiles, and the 2.5th and 97.5th percentile forecasts from the day that the surveillance system has recorded data up to until day 210 (week 30), the end of the observation period.

2.2. Ethics

The Lehigh University Internal Review Board reviewed this experiment and deemed this work exempt (Project: 2014747-1) on 2023-02-01.

2.3. Simulation of incident hospitalizations for experiment

The fictitious epidemic that we generated for the current and two past seasons followed an SEIR model and included an additional compartment to account for the movement of individuals to a hospitalized category which we call a SEIRH model. We presented to the crowd the same true, underlying number of incident hospitalizations, but modified how many days the crowd could observe this time series and the level of noise added to this underlying number of hospitalizations. The same underlying curve allowed us to better compare the performance between members of the crowd that observed different levels of noise, observation time, and whether they received model guidance or not (see Fig. 2 for the three underlying curves that we presented).

The SEIRH model supposes that the proportion of individuals in the susceptible (S), exposed, latent, or pre-infectious period (E), the infectious period (I), removed/recovered (R) and hospitalized category (H) evolves as follows

$$dS/dt = -\beta SI$$

$$dE/dt = \beta SI - \sigma E$$

$$dI/dt = \sigma E - I[\phi\gamma + (1-\phi)\gamma]$$

$$dH/dt = I\phi\gamma - \kappa H$$

$$dR/dt = I(1-\phi)\gamma + \kappa H$$

where β describes the effective contact rate, $1/\sigma$ describes the expected duration that an individual spends in the latent period before progressing to the infectious period, ϕ describes the fraction of individuals that move from the infectious period to hospitalization, $1/\gamma$ describes the duration an individual spends in the infectious period, and $1/\kappa$ describes the amount of time spent in the hospitalized period until moving to the removed/recovered period.

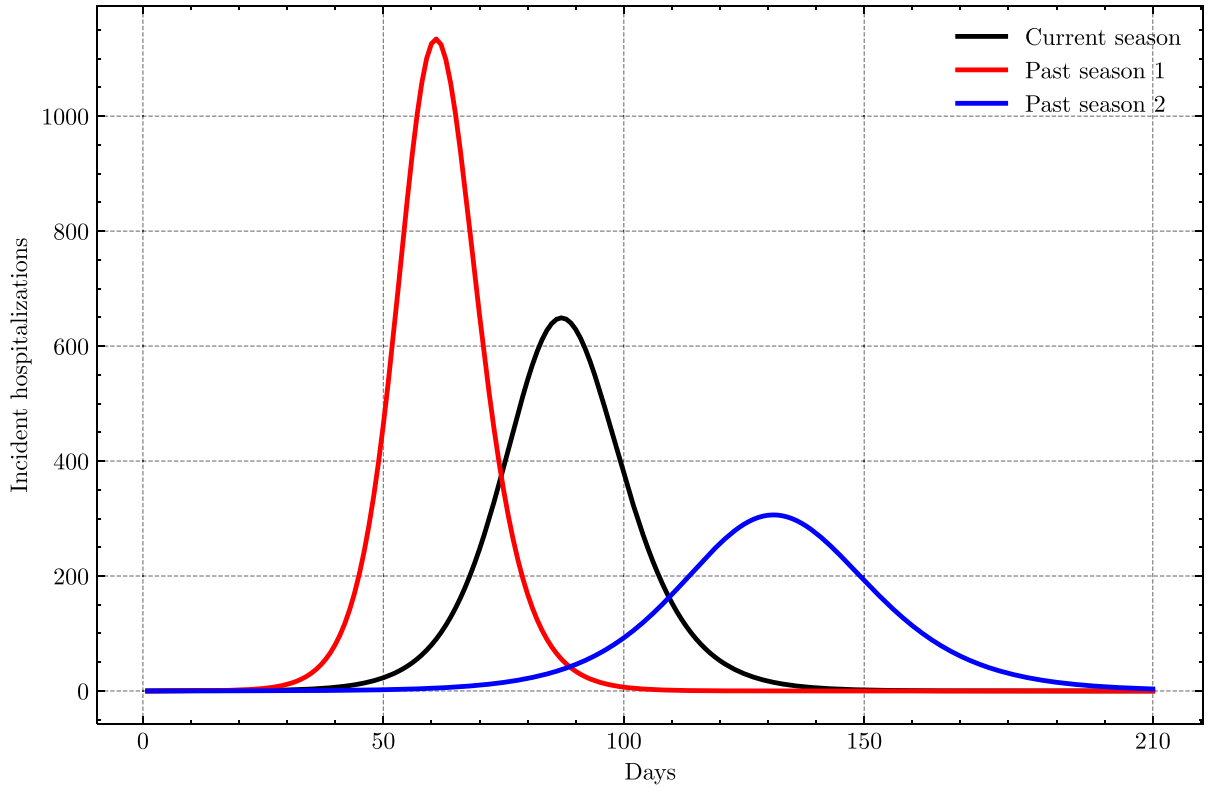


Fig. 2. The underlying number of incident hospitalizations that were presented to the crowd for the current season (black), and two past seasons (blue and red). These three trajectories were kept constant throughout the experiment so that we could better compare model performance across treatments. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

For our simulation, we set the initial conditions for this system to $E_0 = 5/(12 \times 10^6)$; $I_0 = 5/(12 \times 10^6)$; $S_0 = 1 - I_0$; $H_0 = 0$; $R_0 = 1 - (S_0 + E_0 + I_0 + H_0)$ where ρ describes the proportion of individuals who are susceptible at time point zero. The parameter values for our simulation were set to $\beta = 0.058$; $1/\sigma = 2.0$; $\phi = 0.025$; $1/\gamma = 3.0$; $1/\kappa = 7.0$; $\rho = 0.10$. We used Runge–Kutta 4–5 via the odeint module in scipy to solve the above initial value problem (Virtanen et al., 2020). To compute incident hospitalizations $h(t)$ at time unit t from the number of hospitalizations $H(t)$ at t we included a sixth state in the above system of differential equations that keeps track of cumulative incident hospitalizations: $dh/dt = I\phi\gamma$. Incident hospitalizations can then be recovered by computing the difference between successive cumulative incident hospitalizations.

We assume that our surveillance system records a noisy observation of the number of incident hospitalizations at time t according to a negative binomial distribution that depends on the true number of incident hospitalizations at time t and variance v ,

$$\text{Surveillance observation}_t \sim \text{NB2}(h(t), v),$$

where the probability mass function for the negative binomial distribution is parameterized to input the mean and a concentration parameter that adjusts the variance. We used the NegativeBinomial2 class from the numpyro ‘distribution’ method here (Anon, 0000c) which considers the Negative binomial as a Gamma–Poisson compound distribution.

$$X \sim \text{NB2}(\mu, c); f_X(x) = \text{Poisson}(\lambda); \lambda \sim \text{Gamma}(c, \mu/c)$$

$$\text{where the Gamma density is } f(x; \alpha, \beta) = (\alpha^\beta / \Gamma(\beta)) \cdot x^{\alpha-1} e^{-x/\beta}.$$

2.4. Summary of control and chimeric models

All three models are framed as variations on the SEIRH model where the ‘H’ stands for hospitalization. The prior

$$p(\theta, \iota, [\tau, \mu]) \quad (1)$$

can be partitioned into three vectors. $\theta = [\mathcal{R}_0, \sigma, \gamma, \phi, \kappa, \rho]$ contains parameters that control how the infectious agent spreads through the host population: the reproduction number $\mathcal{R}_0$, the rate (σ) of movement from susceptible (S) to exposed (E), the rate (γ) from exposed to infected (I), the proportion (ϕ) of infected who are hospitalized, and the rate (κ) from hospitalized (H) to ‘removed’ (R). In addition this vector includes a parameter ρ that estimates the constant fraction of incident hospitalizations that are observed by the surveillance system. $\iota = [S_0, E_0, I_0, R_0, H_0]$ contains the initial proportions of S , E , I , R , H . Finally, the model above is expanded (see Osthus et al., 2017) to include the peak time (τ) and the peak intensity (μ) of hospitalizations. Because we have expanded the above model to include parameters for peak time and intensity (μ, τ), we can allow human judgment to assign a prior over (μ, τ) which will impact the posterior samples over the number of incident hospitalizations. All models assume that the observational process follows a negative binomial.

The first control model is trained on the current number of incident hospitalizations as captured by this synthetic surveillance system. The second control model first trains on the peak time and intensity for the last and second to last seasons. Training constructs an informed prior over all parameters θ , ι , and $[\mu, \tau]$ which is then used to train on surveillance data for the current season. The chimeric model takes the same approach as the second control model but adds human judgment. The model is first trained on past seasonal peak data (like in the second control model) and a 2D Gaussian kernel (see supplement for example of human judgment Gaussian kernels) that is constructed from human judgment predictions of peak time and intensity. This training develops an informed prior which is then used in training on current season surveillance data.

The in depth methodology for combining surveillance data and human judgment is presented in the supplement under the heading ‘Problem formulation and models’. This section in the supplement also includes methodology for the two control models that we compared to this novel, chimeric forecast.

3. Results

We compared our chimeric model to two control models. One control model learns from current season surveillance data only. The second model learns from current season surveillance data and peak time and intensity from the two past seasons that were presented to the crowd.

We present four key results associated with a model that includes surveillance data and human judgment: (i) including human judgment can better cover the true underlying number of incident hospitalizations in the system, (ii) a model that includes human judgment can produce more accurate estimates of the peak time and peak intensity, (iii) human judgment predictions alone make more accurate predictions of peak time compared to peak intensity, and (iv) humans most often rely on the observed current and past season data than on a model that was trained on this data.

Methodology to evaluate predictions between the two control models and the chimeric model can be found in detail in the supplement section: ‘Evaluating performance of forecasts’. In brief, to evaluate peak day performance we compared posterior samples minus the ground truth peak day for the current season. Differences in posterior minus truth between models were summarized with median, 25th, and 75th percentiles, and formally compared by computing a Kruskal–Wallis test. For peak intensity we computed the probability integral transform (PIT) value as $PIT(f, t) = P(X < t) = F_x(t) = \int_{-\infty}^t f(x); dx$ where f is the predictive density and t is the true peak intensity value. Differences in PIT between models were summarized with median, 25th, and 75th percentiles, and formally compared by computing a Kruskal–Wallis test.

3.1. Highlighting impact of including human judgment

Fig. 3 illustrates the improvement in forecast performance of a chimeric model compared to two control models for predictions of incident hospitalizations at a time when the surveillance system has recorded the number of daily incident hospitalizations up until four weeks and one week before the peak. The SEIRH model trained on current season data (blue) for both four weeks and one week before the true peak generated a 25 percent prediction interval that does not capture the underlying epidemic, and a 50 and 95 percent prediction interval that is too broad (Fig. 3A and 3D). A model incorporating current season data and past peak times and intensities generated prediction intervals that are similar to the model trained on current season data only (Fig. 3B and 3E). A chimeric model that incorporates current season surveillance data, past season peak time and intensities, and crowdsourced human judgment predictions of the peak time and intensity improves upon a model trained on current and past surveillance data. Compared to a model that uses current and past surveillance data, the median prediction from the chimeric model at four weeks before the peak is closer to the truth and prediction intervals are more focused and cover the true epidemic. For one week before the peak, the chimeric model median is closer to the underlying true incident number of hospitalizations and lower bounds of each prediction interval better cover the truth.

3.2. Including human judgment improved predictions of peak time, intensity

From our factorial experiment we find that a chimeric model, compared to models trained only on surveillance data, improves upon predictions of peak time (Fig. 4A–4C) and peak intensity (Fig. 4D–4F; see supplementary table 1 for details on effect estimates and statistical testing). The improvement in forecasts of peak time is observed for varying lengths of time that the surveillance system collected data (Fig. 4A), whether humans received a model to help guide their predictions or not (Fig. 4B), and for high noise in the surveillance system (Fig. 4C). For length of surveillance time, chimeric forecasts are closer, and underestimate the truth for 4 weeks (median [25th, 75th

percentile] diff = -2 [$-13, 7$]) and for 3 weeks (-4 [$-13, 7$]) before the peak compared to the control model trained on surveillance (4 weeks = 3 [$-8, 13$]; 3 weeks = 6 [$-9, 15$]) and the control model trained on surveillance plus past peaks (4 weeks = 3 [$-8, 13$]; 3 weeks = 6 [$-9, 15$]). The comparisons between chimeric and non-chimeric models were significant (p values < 0.01). For the chimeric model, peak timing was not impacted by model guidance (median [25th, 75th percentile] diff = 1 [$-6, 9$]) or no guidance (1 [$-6, 9$]). A chimeric model improved peak time prediction in high noise (median [25, 75] diff = 1 [$-7, 9$]) vs the control model trained on surveillance (4 [$-1, 16$]) and trained on surveillance plus peaks (4 [$-1, 16$]; $p < 0.01$); for medium noise ($p < 0.01$); and for low noise ($p < 0.01$).

A chimeric model, on average, also outperformed models trained on only surveillance data for predictions of peak intensity (Fig. 4D–4F).

The median forecast of peak intensity from a chimeric model, compared to control models, was often closer to true peak intensity at all lengths of surveillance data. Significant differences (according to a Kruskal–Wallis test) in PIT scores were present at 4 and 3 weeks before the peak (4 weeks $p < 0.01$ and 3 weeks $p < 0.01$). Both control models more often over estimated peak intensity (median PIT values for control model less than 0.38 at 4 and 3 weeks). Performance of forecasts of peak intensity for the chimeric model for model guidance (Fig. 4E) was similar when model guidance was present (median [25, 75] PIT = 0.43 [$0.21, 0.50$]) vs absent (0.44 [$0.24, 0.51$]). For levels of noise (Fig. 4F), high noise showed a significant difference ($p=0.05$) between a chimeric model (median [25,75] PIT = 0.45 [$0.30, 0.51$]) and the control model trained on surveillance (0.31 [$0.17, 0.32$]) and surveillance plus past peaks (0.31 [$0.17, 0.32$]). However, chimeric forecasts performed more poorly as the surveillance system collected data at one week before the peak and two weeks after the peak which was similar to the performance of control models (Fig. 4D).

3.3. Humans make more accurate predictions of peak time than intensity

Human judgment predictions of the true peak time (Fig. 5A–5C) were more accurate than predictions of peak intensity (Fig. 5D–5F; see supplementary table 2 for details on effect estimates and statistical testing).

Human predictions of the peak time and peak intensity were worse when the crowd was presented vs not presented with model guidance—a SEIRH model trained on the surveillance data from the current season (Fig. 5B and 5E). Define bias as the average predicted peak (intensity/time) minus the truth. Then, for peak time, the average change in bias between model guidance (5.02 days) and no guidance (0.43 days) was 4.6 days; $95CI=[2.25, 6.94]$; $p<0.01$. For peak intensity the average change in bias between model guidance (535 hospitalizations) and no model guidance (347 hospitalizations) was 188.94 hospitalizations; $95CI=[138.65, 239.23]$; $p<0.01$.

The level of noise did not seem to impact predictions of peak time ($p=0.47$), but increasing levels of noise increased the median difference from the human predicted peak intensity to the truth (low noise median= 450 , medium noise= 460 , high noise = 550 ; $p<0.01$) (Fig. 5C and 5F).

Predictions of peak intensity were worse one week before and two weeks after the underlying peak had passed compared with two through four weeks before the peak (Fig. 5A and 5D; $p<0.01$).

3.4. Self-reported factors human considered when forming a prediction

The majority of humans relied on noisy surveillance data from the current season to form predictions of the peak time and intensity (Fig. 6). In most cases, humans relied more on current season surveillance data as the size of this dataset grew (Fig. 6A). Humans relied on past season data to make predictions but used this information less as more current season data was collected (Figs. 6B and 6C). When we provided a model that was fit to the current surveillance data, approximately 50% of the crowd reported that they considered this model when generating a prediction (Fig. 6D).

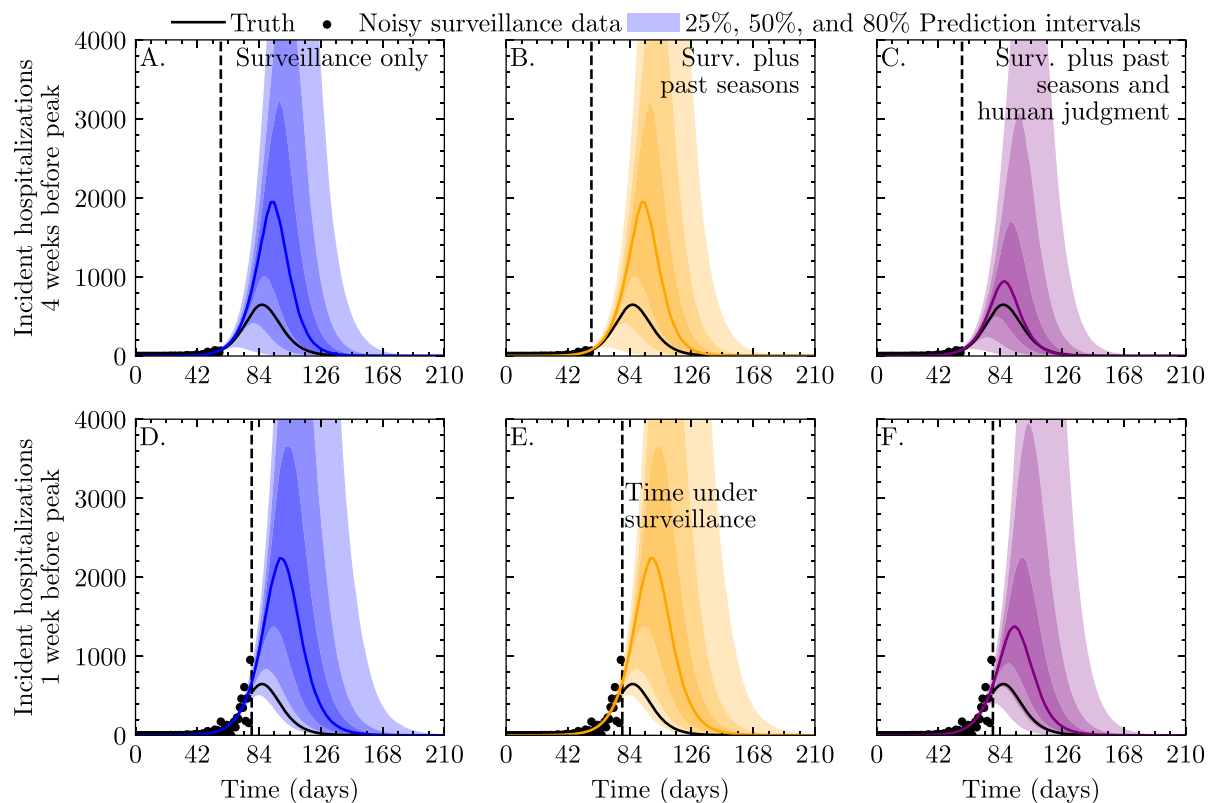


Fig. 3. Forecast performance for two control models and a chimeric model. A median prediction (solid line), 25% prediction interval (PI), 50%, and 80% PIs of incident hospitalizations generated by a fictitious epidemic (black line) and recorded by a noisy surveillance system (black circles) for three models: (A. and D.) a model trained on current season surveillance data only, (B. and E.) model trained on current season data and peak data from the past two seasons, and (C. and F.) a chimeric model that learns from surveillance data, past season peak data, and crowdsourced human judgment predictions of peak time and intensity. The top row presents models trained on data up until four weeks before the peak and the bottom row presents models trained up until one week before the peak.

4. Discussion

Collecting human judgment predictions about peak time and intensity took three days which is fast relative to the typical one week cadence of producing forecasts of the trajectory of infectious agents for platforms such as FluSight (Anon, 0000d). Our results support the hypothesis that a generalist crowd can extract multiple cues from an active surveillance system that is generating noisy observations from a nonlinear, dynamical system and map these cues to a two dimensional vector of peak time and peak intensity.

Performance of humans varied. We hypothesize that choice overload may play an important role in why humans were better able to predict the peak day compared to the peak intensity of the underlying epidemic (Chernev et al., 2015). There was an order of magnitude larger number of choices for peak intensity than for peak time, and past work has shown that choice overload can cause humans to make poor decisions under a diverse range of settings (Chernev et al., 2015). We hypothesize that predictive tasks too may be impacted by choice overload.

In addition, human predictions of peak intensity tended to be higher on average than the true peak. We cannot know for certain the reasons that led to human overestimating peak intensity. Instead, we hypothesize three potential reasons for this bias. Humans may have relied too heavily on past seasonal data when current season data was sparse. The approximate average value of the peak for the past season and peak for the second to last season was 1000 (true peak was 650). A second hypothesis is that human may have overvalued the noisy observed data. At (unknown to the participant) 2 weeks after the peak, the average between the two highest observed values for the current season for the low noise condition is 1100, medium noise 1250, and high noise 1500. We see a similar trend in Fig. 5F. A third, final hypothesis that

we propose is the crowd could have been anchored/primed to choose a value that is in the center of the possible lower and upper bounds. In other words, the participant could have reasoned that the peak should be near 1250, the “middle value” between 0 and 2500.

We suspect that when including for participants a model in addition to the data, the model served as an anchor which caused human judgment predictions of peak intensity to be worse compared to experimental arms when the model was absent. Because participants in this experiment self-reported that they did not often rely on the model to inform their predictions, we suspect that this anchor may be not conscious, or automatic, among participants (Tversky and Kahneman, 1974). This highlights not only the role that a model can play in biasing the collection of predictions, but the importance of how models may frame decision making even when the underlying data is present. Another potential anchor was that the current season peak was (though unknown to participants) between the peak for past season data and peak for the second to last season data.

Previous work in crowdsourced human judgment predictions of infectious disease often focus on aggregating predictions generated by a crowd (Farrow et al., 2017; McAndrew et al., 2022c). Though these works show promise for human judgment, methods that map only human judgment to a sequence of predictions may likely be subject to cognitive biases present in the crowd and so there exists an extensive amount of methodological work which attempts to correct for these biases (Goodwin, 2000; Reimers and Harvey, 2023; Goodwin and Wright, 1993). The chimeric model presented here is closer to a ‘human-in-the-loop’ process (Wu et al., 2022; Zhang et al., 2017). The model first uses human judgment to refine a prior density over parameters that control the evolution of disease states of the model and then generates proposed trajectories of the number of incident hospitalizations that are compared to observed surveillance data. This work could be the

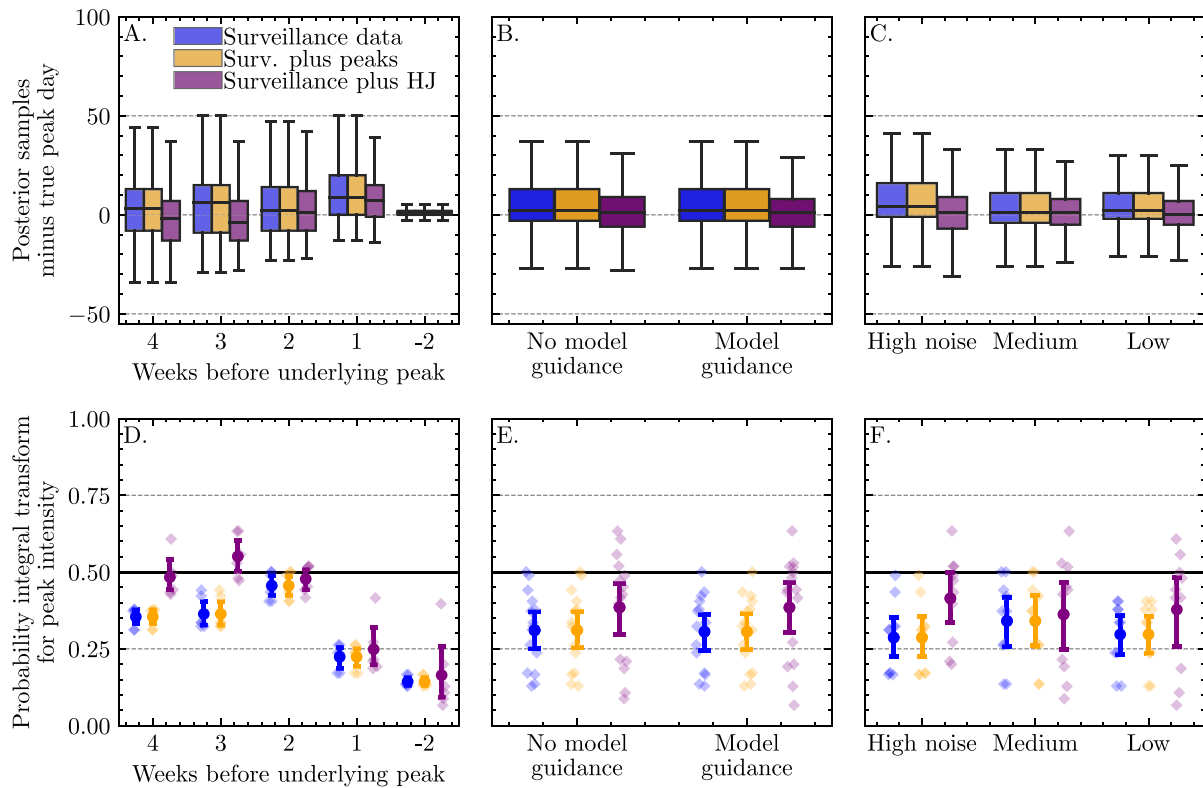


Fig. 4. Predictions of peak time and intensity for control models and a chimeric model. Comparisons between a chimeric (purple) model and two control models (blue and orange) of posterior predictive distributions of the peak time minus the truth stratified by (A.) The length of time the surveillance system was observed, (B.) whether the crowd was provided model guidance or no guidance, and (C.) low, medium, and high noise in the surveillance system. We also present probability integral transform (PIT) values for model forecasts of peak intensity (D.)–(F.). A PIT value of 0.50 indicates that the median prediction was equal to the true peak intensity. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

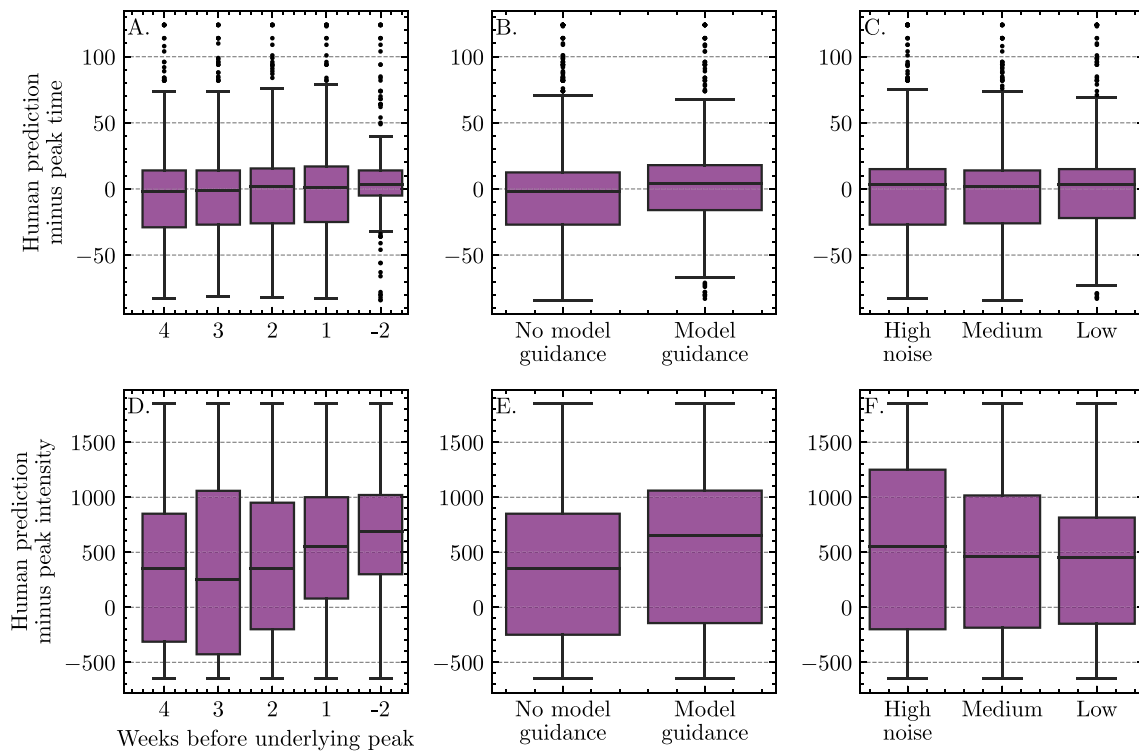


Fig. 5. Human judgment predictive performance of peak time and intensity. Human judgment performance for peak time (A.)–(C.) and peak intensity (D.)–(F.) for different lengths of time the surveillance system has been observed, whether human were presented with an overlay of a model or not, and differing levels of noise in the surveillance system.

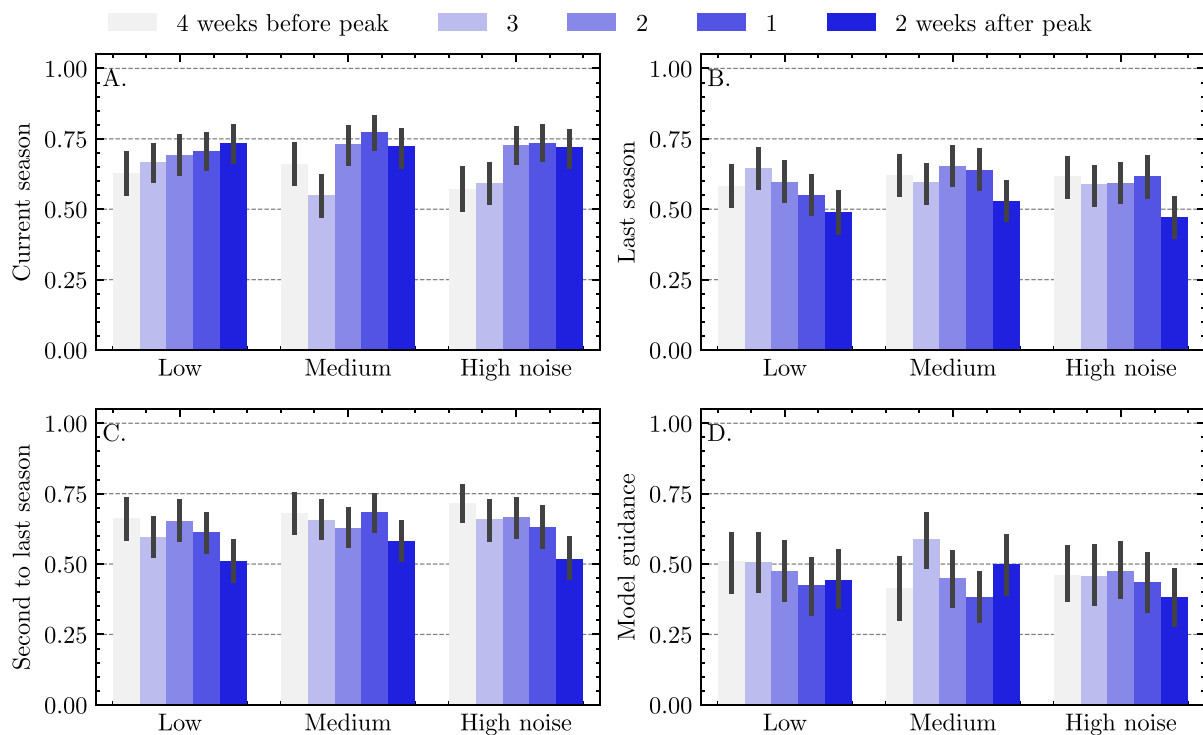


Fig. 6. The proportion of the crowd that considered current season data, last season data, second to last season data, and model guidance when generating predictions of peak time and peak intensity. Proportions are stratified by differing lengths of data presented to the crowd and noise in the surveillance system.

starting point for an interactive style of ‘human-in-the-loop’ machine learning of the evolution of infectious agents where humans are able to suppose specific attributes for how an infectious agent propagates through a host population, visualize a model that is built with these attributes, and then iterate. Work similar to this proposal has already emerged and is called hybrid forecasting, a discipline that focuses on the interaction between machine learning models which generate forecasts and a crowd (Abeliuk et al., 2020).

Past work in human judgment, crowdsourcing, and infectious disease (ID) has focused on prediction of observations produced by a surveillance system. Farrow et al. (2017), McAndrew et al. (2022c), McAndrew and Reich (2022), Bosse et al. (2022), McAndrew et al. (2022b) Little work in ID has tested, in a more controlled fashion, the ability of humans to predict targets associated with the trajectory of an infectious agent where the ground truth is known and hidden from participants. However, this experimental work to determine how to best extract human judgment predictions to improve model performance is in early phase. There is a plethora of work to complete to better understand the impact of human judgment on model performance.

Human judgment predictions of the peak time and intensity could also be viewed as a method for regularizing compartmental models. Our chimeric model trained first on human judgment to build an informative prior, emphasizing specific regions of the parameter space that aligned with human judgment predictions and de-emphasizing regions that did not align. In this sense, the inclusion of a human judgment prior could be recognized as a type of regularizer.

This study was limited by the use of a simulation to measure the ability of human judgment to improve predictions of a computational model and more detailed information about how humans form predictions. Our simulation reduced the complexity of how a real surveillance system collects and disseminates data on the number of incident hospitalizations. In the future we aim to test human judgment predictions of the peak time and intensity of an empirical time series produced by a real surveillance system. Though we collected information about potential factors that humans applied to this predictive task, we did not collect more detailed data about this population which could

include demographic, background information about participants, and eye tracking technology to extract hidden features of the surveillance system that humans use to form predictions. In the future we plan to address the above limitations with additional studies, as well as pose different questions about the trajectory of an infectious agent that may improve forecasts. Our elicitation procedure asked participants to provide a single point prediction. A point prediction prevented participants from expressing uncertainty about their prediction. In the future, several elicitation procedures should be tested and compared to one another.

Chimeric modeling highlights our ability as humans to contribute to the health and well being of others by producing predictions that improve forecasts of the trajectory of an infectious agent. We expect chimeric modeling may play an important role in the future of epidemic forecasting.

Code availability

Human judgment predictions that were collected from Prolific are available at the same GitHub repository https://github.com/computationalUncertaintyLab/hj_guided_prediction in a folder titled ‘data_from_prolific’. Readers will find in this folder a CSV file titled ‘prolific_data.csv’ which is the data that we collected for this study.

CRediT authorship contribution statement

Thomas McAndrew: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing. **Graham C. Gibson:** Conceptualization, Writing – original draft, Writing – review & editing. **David Braun:** Writing – original draft. **Abhishek Srivastava:** Conceptualization, Visualization. **Kate Brown:** Conceptualization, Visualization.

Declaration of competing interest

None

Data availability

All code and data is fully available on GitHub at https://github.com/computationalUncertaintyLab/hj_guided_prediction.

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Appendix A. Supplementary materials

- Simulation of Crowdsourcing experiment
- Problem formulation and models
- Evaluating performance of forecasts
- Survey
- Figures S1 to S5
- References (42–44)

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.epidem.2024.100756>.

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